# Reach Probabilities Explain and Predict Generator Tradeoffs in Stateful Authorization Testing

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***Abstract*—Two kinds of test generators are used against stateful APIs: random generators and generators that mutate valid workflows. Neither wins in general. Workflow mutation finds defects that need several prior steps, and random programs find shallow scope violations faster. We show that this tradeoff can be computed before any campaign runs. For an authored appointment/payment service, we record the exact transition table of the real HTTP/SQL backend over its 175-action campaign domain; from that table a dynamic program, exact with respect to the recorded table, gives each generator's probability of detecting a defect within a call budget. The table is validated three ways: the unchanged campaign code reproduces all 2,740 recorded campaigns field for field; 8,750 transitions reached by real action paths match the table; and a corrupted table fails 19 of 20 replays. With 1,000 seeds per condition, valid workflow banks raise detection of capacity, payment and refund defects from 0–13% to 43–79% at 120 calls, while cross-scope reading moves the other way (99.5% random vs. 30.2% workflow mutation). In a prospective test, predictions derived only from a specification model and frozen by hash reproduce every generator ordering with a predicted gap of at least ten points: 26/26 on 14 systematically generated mutants and 9/9 on OWASP crAPI, a service authored by others, against 450 live campaigns (30 seeds per cell) (the nine crAPI pairs come from three rule families and share the same campaigns). Ordering and probability are distinct results: one of the fifteen crAPI calibration cells deviates at $p < 0.05$ uncorrected (15/30 observed vs. 20.7 predicted; none after Holm correction), yet its ordering still holds, and pooled per generator valid-bank mutation on crAPI is under-predicted (20 observed vs. 12.4 expected), a specification/implementation gap we report. The generator fingerprint differs from the authored service exactly as the model predicts. The contribution is a method that turns generator choice from trial into computation; it is not a new fuzzing algorithm, a security proof, or a claim of superiority over any tool.**

***Index Terms*—software testing, REST API fuzzing, authorization, stateful testing, test generation, reach probability, reproducibility**

## I. Introduction

Authorization in business APIs depends on more than the operation name. It depends on who calls, on which resource, and after which history. A refund can be legitimate once and a violation the second time. A confirmation is legitimate only after payment. A read is legitimate only within the caller's scope.

Test generators for stateful APIs differ in how they reach such histories: random programs sample operations and arguments independently; workflow-seeded generators mutate known valid tasks; and feedback-guided generators keep a corpus of programs indexed by abstract observations [1], [2], [3], [4], [5], [6].

Our earlier evaluations of these generators, over several fixtures and phases, gave a consistent but unexplained pattern. Workflow mutation won on repeated refunds, unpaid confirmation and overbooking. Random programs won on cross-scope reads. Richer feedback helped some defects and hurt others. A list of per-property wins and losses does not tell a practitioner what to run next, and with 10–30 seeds per condition many differences were not statistically resolvable.

This paper replaces that list with a computation. For a finite campaign domain, the probability that a generator detects a defect within a budget is determined by two things: the service's transition structure and the generator's program distribution. Both can be computed exactly. The *minimal witness depth*, the number of calls in the shortest violating history, is a useful proxy for the comparison between random programs and workflow mutation. The quantity that decides the comparison, however, is the generator's *reach probability* for the violating histories.

**Contributions.**

1) *An exact reach analysis for stateful authorization testing.* We record transition tables from the real HTTP/SQL backend. Episode and budget distributions are computed exactly for random, prefix-extension and mutation generators, validated by exact replay of 2,740 recorded campaigns, path-based fidelity checks, and an exhaustive check of the mutation enumeration.
2) *Effect sizes at resolvable precision.* Running the unchanged campaign code on the tables with 1,000 seeds per condition settles contrasts that 30-seed experiments could not; among them, corpus guidance underperforms uniform workflow mutation on history-dependent defects.
3) *A prospective test across the specification/implementation boundary.* Predictions derived only from the specification model order the generators correctly on 14 systematically generated mutants and identify three as undetectable within the bounded

domain; and on OWASP crAPI, an externally authored service.

4) *A public artifact* with tables, frozen predictions, raw campaigns and replay commands.

We do not claim a new fuzzing algorithm, security verification of any production system, or superiority over any tool.

Each result is labeled with one of three tiers: *pipeline validation* (the tables and model reproduce what was recorded), *exploratory* (analysis of designs and defects that informed this work), and *prospective* (predictions frozen before execution on new defects).

## II. Related Work

**Stateful REST testing.** RESTler infers producer–consumer dependencies and uses execution feedback [1]; its security checkers test namespace, hierarchy and use-after-free style rules [3]. EvoMaster generates REST tests with evolutionary search [7]. Sahin, Zhang and Arcuri [6], extending [8], are the closest work to our setting: they add authentication and access-control oracles, some multi-call (one user creates a resource and another probes it), run in a post-processing phase over the tests the search produced, and evaluate one generator on many APIs (including crAPI). They do not compare generators and do not model how likely a generator is to reach a violation, and their oracles do not cover history-dependent business violations such as repeated refunds, confirmation before payment or capacity — which are our focus. RESTTESTGEN [4], mass-assignment testing [5], specification-based access-control detection [9] and Schemathesis [10] add nominal/error scenarios, vulnerability-specific oracles and schema-derived property checks. Our generators and oracles have these antecedents; our contribution is to explain and predict *which* generator detects a given history-dependent defect.

**Probabilistic analyses of testing.** Arcuri, Iqbal and Briand give closed-form results, framed as a coupon-collector problem, for the time random testing needs to reach targets and for how predictable its reruns are [11]; our random-generation reach probability is the same quantity made exact for one concrete stateful service rather than a closed form over targets. Böhme and Paul model testing as sampling from abstract error-based input partitions and derive bounds on the cost per test beyond which random testing is more efficient than systematic partition testing on average [12]. Probabilistic Concurrency Testing defines bug depth and gives detection bounds in terms of depth [13]; our use of minimal witness depth follows that tradition. Sequence covering arrays target $t$-way event orders [14]. Bounded-step probabilistic reachability is itself standard in probabilistic model checking; our use of it is not the novel part. Those analyses work with abstract partitions, targets and parameters, and compute reachability of a property. Our contribution is narrower and empirical: we derive the detection-probability distribution of *concrete* test generators (random, mutation and prefix extension, including their charged initialization cost), over a transition table recorded from a running HTTP/SQL service, and contrast those generators under a shared budget against live campaigns. "Exact" here means exact with respect to that recorded table (the episode pmfs are rationals), not a guarantee about the service beyond the bounded, finite domain.

**Evaluation practice.** We follow guidance on repeated randomized trials and on separating detected inputs from distinct defects [15]. From [16] we adopt the Vargha–Delaney $A_{12}$ effect size and its advice to report raw $p$-values; our paired binary detection counts additionally use an exact McNemar test with Holm-adjusted $p$-values (McNemar suits paired success/failure outcomes, which differ from the continuous performance measures [16] centers on).

**Enabledness abstractions.** Enabledness-preserving abstractions [17], [18], [19] support contract validation and coverage. Our earlier counterexample [20], [21] shows that one particular global enabledness projection merges states that need different authorization decisions; it does not refute those complete methods, and it now appears only as background (Appendix A).

## III. Systems, Oracles and Threat Model

**TurnoSalon** is the primary system, a FastAPI/SQLite service with two salon branches, two customers and two staff members. Its business rules: reservations occupy one unit of slot capacity; payments debit server-priced synthetic credit; confirmation requires payment; cancellation releases capacity; and a staff refund credits the customer exactly once. Identities are resolved from signed sessions, and writes commit in independent SQLite transactions.

A separate specification oracle derives the expected status code, business state and output for every call, then checks invariants over capacity, payment, refunds and ledger balance. The oracle imports no implementation code.

The campaign domain is finite: 7 operations, 5 principals (including an anonymous one) and 5 resource identifiers, giving 175 actions; programs of at most 8 actions; 120 calls per campaign, with 480 as a sensitivity condition. The assumed attacker chooses operations and identifiers with fixture credentials, including foreign identifiers and repeated calls. Out of scope: resource creation, concurrency within campaigns, identity management and server compromise; concurrent contracts are tested separately (Appendix A).

**Reviewed defects.** Four deliberate mutants each omit one guard: booking read scope, slot capacity, payment before confirmation, and the already-refunded check. They were reviewed and used across phases v0.5–v0.8, informed this work, and are therefore exploratory evidence.

**Systematic mutants (prospective).** We parse `SalonService.read` and `SalonService.transition` and remove each top-level disjunct of every `if ...: raise` guard. This yields 18 sites with no hand selection; four coincide with the reviewed defects, and the other 14 form the prospective set. Mutants are compiled in memory and injected at adapter construction, and the measured source is never edited.

**Secondary fixtures.** A synthetic document-sharing service with eleven authored defects (v0.2); an existing

FastAPI/PostgreSQL backend by the same author, with a public source subset and mitigation (v0.3–v0.4); and a scoped RESTler setup pilot (v0.7). They are summarized in Appendix A.

## IV. Generators and Campaigns

All generators share one campaign loop. Before each program, the service is reset to the fixture. Every executed call is charged against the budget. An episode stops at its first oracle violation (for crAPI, which holds several defects at once, only at the target property; Section VII-H); the campaign stops at the first matching violation, or when the budget is exhausted. The generators are: **random** (each program has a uniform length from 1 to 8, each action uniform over the 175-action domain); **valid replay** (one of eight legitimate task programs, unchanged); **valid prefix + random** (`seed_extend`: a legitimate task followed by a uniform random suffix); **mutation of valid bank** (`sequence_uniform`: a uniform source and donor task, then one of six generic operators — field change, insert, duplicate, delete, swap, splice); **mutation of random bank** (the same mutator on a length-matched random bank); **corpus-guided mutation** (`sequence_enabled`, `sequence_enriched`: with probability 0.8 the source comes from a 64-entry FIFO corpus keyed by coarse or enriched abstract transitions and scheduled toward rarely selected keys, otherwise from the valid bank); and **balanced portfolio** (random and valid-bank mutation, alternated by the calls each has consumed).

Every generator except random first executes its bank as a charged bootstrap (24 calls for the valid bank). The corpus engines therefore differ from uniform mutation in source selection, eviction and scheduling, not only in their feedback key.

## V. Exact Reach Analysis

### A. Recorded transition tables

For each service variant we run a breadth-first exploration. From each state, every one of the 175 actions is executed through the real HTTP application and SQL database, and we record the response, the next business snapshot and the oracle verdict. States are written into a disposable database before each action. Only states reached through violation-free transitions are expanded, because episodes stop at the first violation; levels 0–7 are expanded, covering every program of up to eight actions. For the corrected service this reaches 723 expandable states and 126,525 recorded transitions; exploration takes about 70 seconds per variant on 14 processes, and parallel and sequential exploration produce identical tables.

### B. Episode and budget distributions

Let $\mathcal{A}$ be the finite action set ($N = |\mathcal{A}|$, here 175). The recorded table gives, for a state $s$ and action $a$, the successor $\delta(s,a)$ and the oracle verdict; write $T(s,a) = 1$ when $a$ triggers the target property at $s$, and call $(s,a)$ *clean* when it produces no stopping violation. An episode executes a program and stops at its first target hit; a non-target violation stops the episode in the salon single-defect setting and is transparent under the crAPI target-only rule (Section VII-H). Both are instances of the same recurrence.

We compute, per generator, the episode pmf $(d_c, r_c)$, where $d_c$ is the probability that the episode's first target hit is at call $c$ and $r_c$ the probability that it ends (stop or program end) after $c$ calls with no target hit. Each generator produces programs differently, so the pmf is computed differently; all three use exact rational arithmetic.

*Random.* Each call is uniform over $\mathcal{A}$ independently of the state, so a forward dynamic program over states suffices. Let $m_c(s)$ be the probability of being at $s$ after $c$ clean calls (no target hit); with $m_0$ the start distribution,

$$d_{c+1} = \tfrac{1}{N} \sum_s m_c(s) \sum_{a \in \mathcal{A}} T(s,a), \tag{1}$$

$$m_{c+1}(s') = \tfrac{1}{N} \sum_s \big|\{a : \ \delta(s,a) = s', \ (s,a) \text{ clean}\}\big| \, m_c(s). \tag{2}$$

The uniform program length $1$–$L$ is handled by averaging the truncations.

*Mutation.* The program is not Markov in the backend state: it is drawn as a whole. We enumerate the mutation operator's entire output — over every source/donor pair from the bank and every branch of the operator — as a set of (program, probability) pairs, run each program deterministically through the table to its first target hit, and add its weight to the corresponding $d_c$ or $r_c$. A unit test drives the real mutator through every random-choice path and checks that this enumeration reproduces its distribution exactly.

*Prefix extension.* The deterministic valid prefix is run through the table by position; the uniform random suffix that follows is then handled by the state dynamic program of (1)–(2) started from the prefix's end state.

**Budget distribution.** A campaign first runs its deterministic charged bootstrap (every generator but random executes its bank; for the valid bank $b = 24$ calls). The bootstrap programs are run through the table in order: if one hits the target at cumulative call $t$, that is a deterministic detection, $F_t = 1$, and the campaign is done; otherwise the bootstrap consumes $b$ calls with no detection. This case is not vacuous — a random bank, for instance, detects cross-scope reads during the bootstrap. After a non-detecting bootstrap, i.i.d. episodes follow. Let $B_n$ be the probability that a fresh episode begins after $n$ consumed calls ($B_b = 1$) and $F_t$ the probability of the first detection at call $t$. For $n + c \le T$,

$$F_{n+c} \mathrel{+}= B_n \, d_c, \qquad B_{n+c} \mathrel{+}= B_n \, r_c, \tag{3}$$

and $P(\text{detect within budget } T) = \sum_{t \le T} F_t$, with expected capped effort $\sum_t t \, F_t + T\left(1 - \sum_t F_t\right)$. The episode pmfs are exact rationals; the budget recurrence (3) is evaluated in floating point. This is exact for *random*, *mutation* and *prefix extension*; the corpus and portfolio engines, whose episodes are not i.i.d., are measured on the tables instead (Section VI).

**Worked example.** Take cross-scope read (depth 1) under *random*. From the fixture state, 8 of the 175 actions are a

TABLE I
SHORTEST WITNESSES FOR THE REVIEWED DEFECTS.

| Defect | Depth | Distinct shortest witnesses |
|---|---|---|
| Cross-scope read | 1 | 8 |
| Overbooking | 2 | 16 |
| Unpaid confirmation | 2 | 8 |
| Repeated refund | 5 | 16 |
| Corrected service | — | 0 |

scope violation, so $d_1 = 8/175 \approx 0.046$ and, over the uniform program length 1–8, the per-episode detection is $\approx 0.185$. Feeding $d_c, r_c$ into (3) with budget $T = 120$ gives $P(\text{detect}) = 0.9964$ and expected capped effort 21.8 calls — matching the observed $99.5\%$ (Table III) without running a campaign.

### C. Validation (pipeline tier)

**Exact replay.** We ran the unchanged v0.5, v0.6 and v0.8 campaign code on the tables. All 2,740 recorded campaigns, across eight generators, five variants and two budgets, matched their archives in all 19 recorded fields (complete traces, responses, minimized witnesses and corpus statistics). As a negative control, a table with one corrupted status code fails 19 of 20 replays. **Path fidelity.** For 10 sampled states per variant we reset the service, executed the real action path leading to the state (no restoration), and compared all 175 transitions against the table: 8,750 of 8,750 matched. **Consistency of predictions.** Across 48 recorded cells (study $\times$ generator $\times$ defect $\times$ budget), observed detections are consistent with the exact probabilities: the minimum two-sided binomial $p$ is 0.07, and 31 cells are informative ($0.02 < p < 0.98$). This is expected and confirms the pipeline; it is not independent predictive evidence. The surrogate runs of Section VI are a different thing: they execute the unchanged campaign code on the validated tables, which reproduce the backend field for field, so they are measurements of generator behavior on the service, not predictions.

### D. Shortest witnesses and reach

For each variant, breadth-first search over clean transitions gives the shortest violating history and the number of distinct shortest witnesses; this depth does not depend on any campaign and, for the four reviewed defects, equals the length of every deletion-minimized witness recorded in v0.5–v0.8 (Table I).

Depth explains why random generation fails on deep defects: a uniform random program contains a specific five-action history with probability on the order of $175^{-5}$. Depth does not, by itself, rank generators that share structure with the witness; the relevant quantity is each generator's per-episode reach probability.

## VI. EXPERIMENTAL DESIGN

Table II summarizes the data and purpose of each tier. The surrogate runs use the original campaign implementations; the table adapter falls back to the real backend for any state outside the table, and no fallback occurred (0 misses).

TABLE II
EXPERIMENTAL TIERS.

| Tier | Data | Purpose |
|---|---|---|
| Pipeline | 2,740 recorded campaigns (v0.5/6/8); 8,750 path checks | Tables and model reproduce recordings |
| Exploratory | Campaign code on tables, 1,000 fresh seeds/cell | Effect sizes for all generators |
| Exploratory | Planned v0.6/v0.8 contrasts, 30 seeds (McNemar, $A_{12}$, Holm) | Record of planned tests |
| Exploratory | Document benchmark, Monte Carlo, 10,000 episodes/cell | Second fixture for reach |
| Prospective | 14 systematic mutants, frozen predictions, 1,260 campaigns | Order and calibration on new defects |

## VII. RESULTS

### A. Initial knowledge and perturbation (exploratory)

Table III reports detection within 120 calls, 1,000 seeds per cell. With the mutator held fixed, a valid bank in place of a length-matched random bank raises detection of the three history-dependent defects by 43–76 points and lowers cross-scope reading by 45 points. Reach explains both directions: valid tasks place the payment, cancellation and refund prefixes one mutation from the witness, but never pair a principal with a foreign booking, so a scope witness needs a field mutation the random generator gets for free on every call. Appending random suffixes to valid prefixes recovers most of the scope detection but little of the history detection; replay without perturbation detects nothing.

The planned 30-seed contrasts point the same way. After Holm adjustment within each study's family (24 in v0.6, 16 in v0.8), all valid-versus-random-bank contrasts remain significant (exact McNemar, Holm $p \leq 0.025$); $A_{12}$ on capped calls is 0.15 (scope read), 0.84 (overbooking), 0.77 (unpaid confirmation) and 0.67 (repeated refund); values above 0.5 favour the valid bank.

### B. Corpus guidance and feedback keys (exploratory)

Table IV gives paired differences from uniform valid-bank mutation, in percentage points with 95% intervals over the same 1,000 seeds. In this configuration corpus guidance never improves on uniform mutation of the valid bank, and on repeated refunds the deficit grows with the budget. The contrast changes source-selection probability, FIFO eviction and rare-key scheduling together, so it does not isolate the value of feedback in general. Holding everything else fixed, the enriched key beats the coarse key only on overbooking ($+20.7 \pm 3.6$ points at 120 calls); on the other three defects the 95% intervals include zero. With 30 seeds the overbooking contrast had raw $p = 0.0063$ and Holm-adjusted $p = 0.063$ — a matter of power, not evidence of no effect.

**Bank sensitivity.** The valid-bank advantage is not a property of "valid tasks" in the abstract; it is a property of whether the bank contains a task whose prefix reaches the witness. Recomputing the exact 120-call order with one task template

TABLE III
DETECTION WITHIN 120 CALLS, 1,000 SEEDS PER CELL.

| Generator | Cross-scope read | Overbooking | Unpaid confirmation | Repeated refund |
|---|---|---|---|---|
| Random | 99.5% | 13.1% | 6.2% | 0.0% |
| Random after charged warmup | 98.8% | 11.7% | 4.7% | 0.0% |
| Valid replay | 0.0% | 0.0% | 0.0% | 0.0% |
| Valid prefix + random | 88.2% | 14.2% | 7.3% | 5.8% |
| Mutation of random bank | 75.2% | 3.2% | 1.5% | 0.0% |
| Mutation of valid bank | 30.2% | 78.7% | 73.0% | 42.9% |
| Corpus-guided, coarse key | 29.1% | 45.8% | 49.3% | 22.0% |
| Corpus-guided, enriched key | 28.1% | 66.5% | 47.3% | 23.5% |

TABLE IV
CORPUS ENGINES MINUS UNIFORM VALID-BANK MUTATION (PERCENTAGE POINTS, 95% INTERVAL).

| Budget | Engine | Read | Overbooking | Unpaid confirmation | Refund |
|---|---|---|---|---|---|
| 120 | Coarse corpus | $-1.1 \pm 3.8$ | $-32.9 \pm 3.9$ | $-23.7 \pm 4.1$ | $-20.9 \pm 3.9$ |
| 120 | Enriched corpus | $-2.1 \pm 3.8$ | $-12.2 \pm 3.8$ | $-25.7 \pm 4.0$ | $-19.4 \pm 3.9$ |
| 480 | Coarse corpus | $-5.3 \pm 3.4$ | $-4.7 \pm 1.3$ | $-5.2 \pm 1.4$ | $-33.6 \pm 3.4$ |
| 480 | Enriched corpus | $-7.2 \pm 3.5$ | $-1.8 \pm 0.8$ | $-6.5 \pm 1.6$ | $-32.7 \pm 3.5$ |

TABLE V
BANK SENSITIVITY: EFFECT OF REMOVING THE ENABLING TEMPLATE.

| Defect | Full bank | After removing the enabling template |
|---|---|---|
| Cross-scope read | random wins | stable under every removal |
| Overbooking | mutation 0.80 | drop rebook task $\rightarrow$ 0.03; prefix extension wins |
| Unpaid confirmation | mutation 0.71 | drop confirm task $\rightarrow$ 0.01; prefix extension wins |
| Repeated refund | mutation 0.42 | drop refund task $\rightarrow$ 0.00 |

TABLE VI
DETECTION AT 120 CALLS: PORTFOLIO VS. COMPONENTS.

| Defect | Portfolio | Random | Valid-bank mutation |
|---|---|---|---|
| Cross-scope read | 93.7% | 99.5% | 30.2% |
| Overbooking | 48.6% | 13.1% | 78.7% |
| Unpaid confirmation | 39.9% | 6.2% | 73.0% |
| Repeated refund | 16.7% | 0.0% | 42.9% |

removed makes this precise (Table V). The depth-1 scope read is bank-independent; for each history defect, removing the single template that supplies the witness prefix collapses mutation's detection and flips the winner. The reach model predicts this: mutation only wins when a bank task sits one operator from the witness. This exposes the dependence on a hand-chosen bank and shows the model explains it rather than assuming it.

### C. Budget and portfolio (exploratory)

At 120 calls the balanced portfolio lies between its components on every defect (Table VI); at 480 calls it reaches 94–100% on the first three defects and 72.2% on refund, against 92.5% for mutation alone. The exact model gives the whole detection curve, not only two budgets (Fig. 1). The observed 1,000-seed fractions fall on the predicted curves, and the ranking is stable away from the smallest budgets: random dominates the depth-1 scope read at every budget, while on the history-dependent defects valid-bank mutation overtakes random after an initial bootstrap region and then stays ahead — the crossover is at 26 calls for unpaid confirmation, 27 for overbooking and 29 for repeated refund, below which random (which needs no bootstrap) is ahead. So beyond about 30 calls a practitioner need not guess where the ranking flips; within the first $\sim$30 it still matters. These values also explain an apparent inconsistency between phases: valid-bank mutation detected cross-scope reads in 8/30 campaigns in v0.6 and 14/30 in v0.8 under identical settings, and the exact probability is 0.30, so the v0.8 result was a favorable draw.

### D. Second fixture: the document benchmark (exploratory)

The document service has a larger action domain (recipients and approval tokens), and its legacy random generator differs from the business generator. We estimated per-episode detection probabilities with the unchanged components, over 10,000 episodes per defect and generator. Minimal witness depth separates the extremes: sharing overwrite (depth 3) and approval replay (depth 5) are found by valid-bank mutation in 10/10 runs, while random finds them in at most 1/10 (per-episode reach $\leq$ 0.0001 vs. $\approx$ 0.026 for mutation); unauthorized update (depth 1) has reach 0.348 for random against 0.003 for mutation (observed 10/10 vs. 6/10). Depth fails for wrong-recipient acceptance: it has depth 2, yet random detected it in 8/10 runs and valid-bank mutation in 4/10, and the per-episode reach ordering agrees with the observed direction (0.0083 vs. 0.0017). The recorded minimal witnesses are an initiated share followed by acceptance from a principal other than the pending recipient; from the valid bank either form needs one specific field change in one of the two sharing tasks. The budget-level predicted gap is small (7.1 vs. 6.3

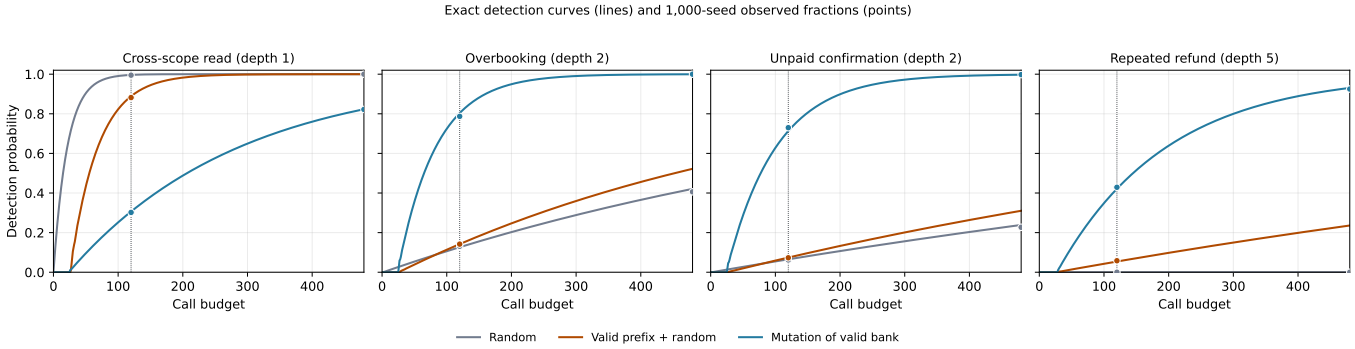



Fig. 1. Exact detection probability versus call budget (lines) with 1,000-seed observed fractions (points) for the three main generators on the four reviewed defects. The dotted line marks the 120-call budget.

expected detections out of 10) and $n = 10$ cannot confirm the direction: this is post hoc evidence that depth is only a proxy, not proof that reach is superior.

### *E. Prospective test on systematic mutants*

The 14 prospective predictions were derived only from the specification model: for each, the corresponding disjunct was removed from the oracle's expected-behavior function, restricted to the operations the implementation guard covers. Before freezing, the specification model was checked on the corrected service and the four reviewed defects; in all five cases its state graph and violation sets were identical to the recorded implementation tables. The predictions were committed by hash before any campaign ran. A first execution exposed a harness defect (the compiled mutant had its own copy of the exception class, turning every rejection into an HTTP 500); that run is preserved and documented, the fix changed neither predictions nor analysis plan, and the injection is now checked against the four known tables before any run.

Results of the second execution (Fig. 2): **Generator order** — all 26 pairs whose frozen probabilities differ by at least 0.1 were ordered correctly by the observed counts. A depth-only rule (depth $1 \rightarrow$ random beats workflow generators; depth $\geq 2 \rightarrow$ workflow generators beat random) also gets the 15 random-versus-workflow pairs it can order, but the remaining 11 pairs compare prefix extension with valid-bank mutation — depth cannot order them, and the exact model orders all 11 correctly. **Bounded-equivalent mutants** — three mutants have no violating program within eight actions; for each, the implementation table is identical to the specification table and neither contains a violation, and they produced 0 detections in 270 campaigns. **Calibration** — the frozen metric is the number of cells with $p < 0.05$ over all cells: 4 of 42 (about 2 expected by chance, so within sampling variation). Restricting to the 20 cells with a non-degenerate prediction, a declared secondary view, leaves 3. The four deviating sites were diagnosed, not assumed to be noise: their implementation and specification graphs are identical, all recorded campaigns replay exactly on the implementation tables, and re-running each deviating cell with 3,000–12,000 fresh seeds matches the frozen probabilities. We found no model error; every deviation is a low-probability draw at $n = 30$. No cell remains significant after Holm correction across the 42 (minimum adjusted $p = 0.22$), and pooled per generator the observed totals match the frozen expectations: random 176 vs. 176.0, prefix extension 160 vs. 160.2, valid-bank mutation 200 vs. 193.2.

### *F. Bounded exhaustive check of the corrected service*

For the corrected service, the table covers every program of up to eight actions over the 175-action domain, and no transition from any reachable clean state produces an oracle violation, so no such program violates the oracle. The scope is exact with respect to the recorded table — this fixture, no resource creation, at most eight actions, and only oracle-observable properties — and the table equals the backend under the assumption that behavior depends only on the observable snapshot (Section IX), checked on a 50-state path-fidelity sample.

### *G. Earlier phases*

The earlier phases remain consistent with the reach account (Appendix A). In the AoristoTech HTTP/PostgreSQL case, valid-task mutation detected repeated cancellation (depth 2) in 10/10 runs against 0/10 for random, while random detected foreign-store reads (depth 1) in 9/10 against 3/10. The RESTler setup pilot could not reach the legitimate refund workflow within its budget, consistent with (though not computed as) the near-zero reach of generators that cannot assemble the payment and cancellation prefix.

### *H. Independent service: OWASP crAPI (prospective)*

The strongest test of the method is a service this project did not author. OWASP crAPI is an intentionally vulnerable application; we study its shop (orders, returns, refunds) at pinned revision `700f03d1`, with every container pinned by digest. We write a specification of its intended behavior from the documented challenges and the OpenAPI descriptions (disclosing that the vulnerable view code was read first), and define a finite domain of two customers and seven shop operations. Because crAPI contains several real defects at

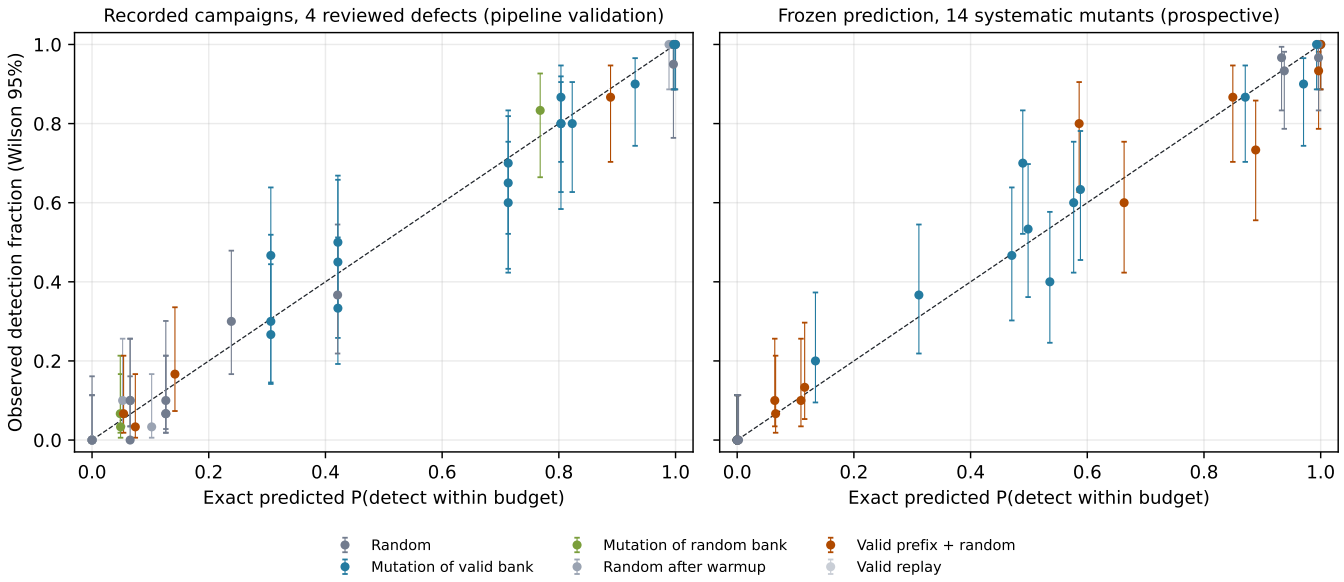



Fig. 2. Exact predicted detection probability versus observed fraction with Wilson 95% intervals. Left: recorded campaigns on the four reviewed defects (pipeline validation). Right: frozen specification-derived predictions for 14 systematic mutants (prospective).

once, we target five rule families and stop each episode only at its target property. The exact reach model is computed per target from the specification, and predictions were frozen by hash before any live campaign ran. We then ran 450 live HTTP/SQL campaigns (3 generators $\times$ 5 targets $\times$ 30 seeds, budget 60) against the unmodified crAPI shop, with the business oracle judging every transition. The only environment change is pointing crAPI's unused external payment gateway at a dead port so order reads do not stall; it never affects the oracle.

The frozen predictions reproduce all nine generator orderings whose probabilities differ by at least 0.1 (Table VII). The return-rules family has no reachable violation within the domain and yields 0 of 30 for every generator, as predicted; refund-beyond-payment is reachable only at depth 3 (predicted $\leq 0.01$) and was hit once, by random (1/30). One of the fifteen cells deviates from its predicted detection at $p < 0.05$ (order-privacy under prefix extension, 15/30 observed vs. 20.7 predicted, $p = 0.029$); with fifteen cells about one such deviation is expected by chance, and it does not change any ordering. The generator fingerprint differs from the authored salon service, and the model predicted the difference: crAPI's defects are shallow (reading another user's order is depth 1; tampering with status or quantity is depth 2), so random and valid-prefix extension beat valid-bank mutation on all three detected families — the reverse of the salon history-dependent defects. Order-privacy (depth 1) is won by random (28/30); order tampering and refund-without-return (depth 2) by valid-prefix extension (24/30 and 16/30); valid-bank mutation is weakest on each (9, 8 and 3 of 30), because crAPI's valid tasks do not sit one mutation from these witnesses. This is prediction across the specification/implementation boundary on a service of another author; crAPI remains an intentionally vulnerable training application with public challenges, so this is implementation independence, not discovery of unknown defects.

crAPI is also the clearest case for computing probabilities rather than using depth as a rule. Applying the paper's own depth heuristic to Table VII — random for depth 1, workflow generators for depth $\geq 2$ — orders 4 of the 6 random-versus-workflow pairs correctly. It errs on both random-versus-mutation pairs at depth 2, where valid-bank mutation is unexpectedly weak on crAPI's shallow defects (tampering 8/30 and refund-without-return 3/30, below random's 13/30 and 8/30), and it has no criterion for the three prefix-extension-versus-mutation pairs. The exact reach model orders all nine. Depth sorts the broad random-vs-history contrast; reach is needed for the rest.

**Calibration on crAPI.** No cell is significant after Holm correction across the fifteen (minimum adjusted $p = 0.44$). Pooled per generator, random (50 observed vs. 45.9 expected) and prefix extension (55 vs. 52.9) are calibrated, but valid-bank mutation is under-predicted: 20 detections against 12.4 expected (exact Poisson-binomial $P(X \geq 20) = 0.019$; 0.058 after Holm across the three generators). Running the unchanged campaign code on the specification model, 2,000 seeds per cell, reproduces the frozen probabilities within 0.02 for every cell, so the dynamic program is not at fault: the gap lies between the specification and the live implementation for mutated programs. Locating it needs the same mutated programs run on both, which we leave as future work; it changes no ordering, but it is the clearest evidence in this paper that ordering is predicted more robustly than probability.

TABLE VII
CRAPI: FROZEN PREDICTED DETECTION PROBABILITY VS. OBSERVED DETECTIONS OUT OF 30, BY GENERATOR. EPISODES STOP ONLY AT THE TARGET PROPERTY (SECTION V-B). ALL NINE SEPARATED ORDERINGS HOLD; ONE OF FIFTEEN CELLS DEVIATES AT $p < 0.05$ ($*$).

| | Random | | Prefix ext. | | Mutation | |
|---|---|---|---|---|---|---|
| Rule family | pred | obs | pred | obs | pred | obs |
| Order privacy (d1) | 0.87 | 28 | 0.69 | 15* | 0.19 | 9 |
| Order tampering (d2) | 0.42 | 13 | 0.68 | 24 | 0.15 | 8 |
| Refund w/o return (d2) | 0.24 | 8 | 0.39 | 16 | 0.08 | 3 |
| Refund > payment (d3) | 0.01 | 1 | 0.01 | 0 | 0.00 | 0 |
| Return rules (control) | 0.00 | 0 | 0.00 | 0 | 0.00 | 0 |

## VIII. DISCUSSION

**A precomputation before fuzzing.** When the action domain is finite and the specification is executable, a generator's detection probability for a hypothesized defect can be computed before running it — about one minute per mutant here. The practical use is allocation: reach probabilities tell where random programs suffice (shallow scope checks), where workflow seeds are necessary (payment and refund histories), and where neither will succeed within a budget. Systematic guard-removal mutants of the specification can stand in for unknown defects.

**Seeds are knowledge, and their placement matters.** The same valid tasks help when mutated internally and barely help as a prefix to random suffixes. Mutation keeps the dependency structure of the task while changing the critical step; a random suffix rarely produces the specific repeated or out-of-order call.

**Corpus guidance needs evidence.** In this domain a coverage-style corpus diverts effort away from the complete valid tasks whose single mutations reach the violations; because the gap persists at four times the budget (large on refunds, smaller elsewhere) it is not only a warmup cost. Other domains may favor corpus guidance. The broader point is that a 30-seed experiment cannot resolve such differences, and an exact replica can.

**Why exact rather than bounded.** Analytic bounds such as PCT's [13] guarantee a minimum detection probability as a function of depth. Our domain is small enough to compute exact values, and exactness is what makes generators with the same depth comparable. The cost is scale: the approach needs a finite action domain, resettable state and tractable reachable state counts (723 here); abstraction or sampling would be required beyond that.

## IX. THREATS TO VALIDITY

**Construct.** Detection is measured against our specification oracle, and the specification and implementation share an author; their identical state graphs on the known variants show agreement, not independent correctness. The prospective predictions come from the specification model, and the specification mutants are mapped to implementation mutants by a hand-written table written with both in view. Graph identity was verified for 12 variants (5 pre-freeze checks and the 7 diagnosed mutants), not for the other 7 prospective mutants; where the graphs coincide, the prospective test measures the fidelity of that mapping as much as a prediction crossing the specification/implementation boundary. An independently authored service, where no such mapping is possible, is the real test; crAPI is a step toward it. Reach probabilities are defined for the campaign domain; arguments outside it (new resources, free-form payloads) are not modeled.

**Internal.** The tables assume behavior depends only on the observable business snapshot, because a state is reached by restoring that snapshot with SQL. Path-based fidelity checks support this on 50 sampled states, not exhaustively. The assumption holds here because booking identifiers are fixed, no resources are created, and execution is single-threaded; it would break under concurrency, state kept outside the database, timestamps, or identifiers whose value changes behavior. In crAPI the risk is higher: sequences are not reset between the service's own calls and authentication carries state, so the symbolic order binding is sound only while behavior does not depend on the concrete order id. We have not verified this directly on crAPI: concrete ids change in every live episode, so a dependence on them would surface as unexplained deviation (a candidate cause of the mutation gap in Section VII-H) rather than be ruled out. The four reviewed defects shaped the method; only the 14 systematic mutants are prospective. The valid-task bank is eight hand-written programs; because the reserve/pay/cancel/refund template is exactly the repeated-refund witness prefix, mutation is one insertion from that witness, and the bank-sensitivity analysis recomputes the order with templates removed or added.

**External.** The authored systems are small and the mutant operator family (guard-disjunct removal) is narrow, so real defects may be distributed differently. The independent evaluation on OWASP crAPI removes the shared-authorship objection for the implementation, but crAPI is still an intentionally vulnerable training application whose defects are public, not an unseen-defect holdout, and its specification was written by us.

**Conclusion.** Surrogate effect sizes use 1,000 seeds per cell within one service and describe search-seed variation, not variation across systems. The 30-seed planned tests are reported with Holm adjustment as the original record, not the basis for the effect sizes above.

### A. Future work

Two external-tool comparisons are left as future work. First, a stronger RESTler configuration, with a dictionary and authentication tuned until it reaches the legitimate payment-and-cancellation workflow, so its detection can be compared under a matched budget rather than the setup pilot. Second, a comparison with EvoMaster and its access-policy oracles [6], [8] on the salon and crAPI services, the closest antecedent to our setting. Both are infrastructure-heavy and neither would license a claim of general superiority: the contribution here is predicting which generator detects which defect, not outperforming a particular tool. A held-out service with unknown

TABLE VIII
EARLIER PHASES.

| Phase | Campaigns | Role here |
|---|---|---|
| v0.1–v0.2 | 1,080 | Document service; second fixture for reach; enabledness counterexample |
| v0.3–v0.4 | 120+ | AoristoTech; depth pattern in an existing backend |
| v0.5 | 640 | TurnoSalon; recorded campaigns replayed exactly |
| v0.6 | 1,200 | Component study; planned contrasts |
| v0.7 | 60 | RESTler setup pilot; no refund reachability |
| v0.8 | 900 | Fixed portfolio, budgets 120/480 |

defects, and extension beyond finite action domains via sampled rather than exact reach, are the other open directions.

## X. CONCLUSION

Generator tradeoffs in stateful authorization testing need not be found by trial. For a finite domain, a generator's probability of exposing a history-dependent violation is computable from the transition structure and the generator's program distribution. Minimal witness depth is a useful proxy for that probability; the probability itself explains the cases where depth fails, and, computed from a specification alone, it predicted the generator order for new systematic defects and for an externally authored service before execution. The same replica turns underpowered comparisons into precise ones: corpus guidance in this service trails plain mutation of valid workflows. The method turns generator choice from a lottery into a computation.

## ARTIFACT AVAILABILITY

Source, tables, raw campaigns, frozen predictions and figures are at https://github.com/PeyranoDev/semantic-gateway-lab. The repository documents the reproduction commands and maps claims to evidence.

## APPENDIX A
## EARLIER PHASES

Table VIII lists the earlier phases and their role here. The enabledness counterexample: let $\alpha_E$ map a state to the set of tool names enabled by existential lifecycle conditions; if two states have equal $\alpha_E$ but different authorization decisions for the same fully specified invocation, then no function of $(\alpha_E(s), a)$ reproduces the decision. Executable witnesses exist in the document service (one sharing prefix designates Bob, the other Mallory) and in the salon service (confirmation of the same booking). The feedback keys are used for search only, while the runtime and oracle inspect full state.

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